Python in the Advanced Labs

https://physicslabs.ucd.ie

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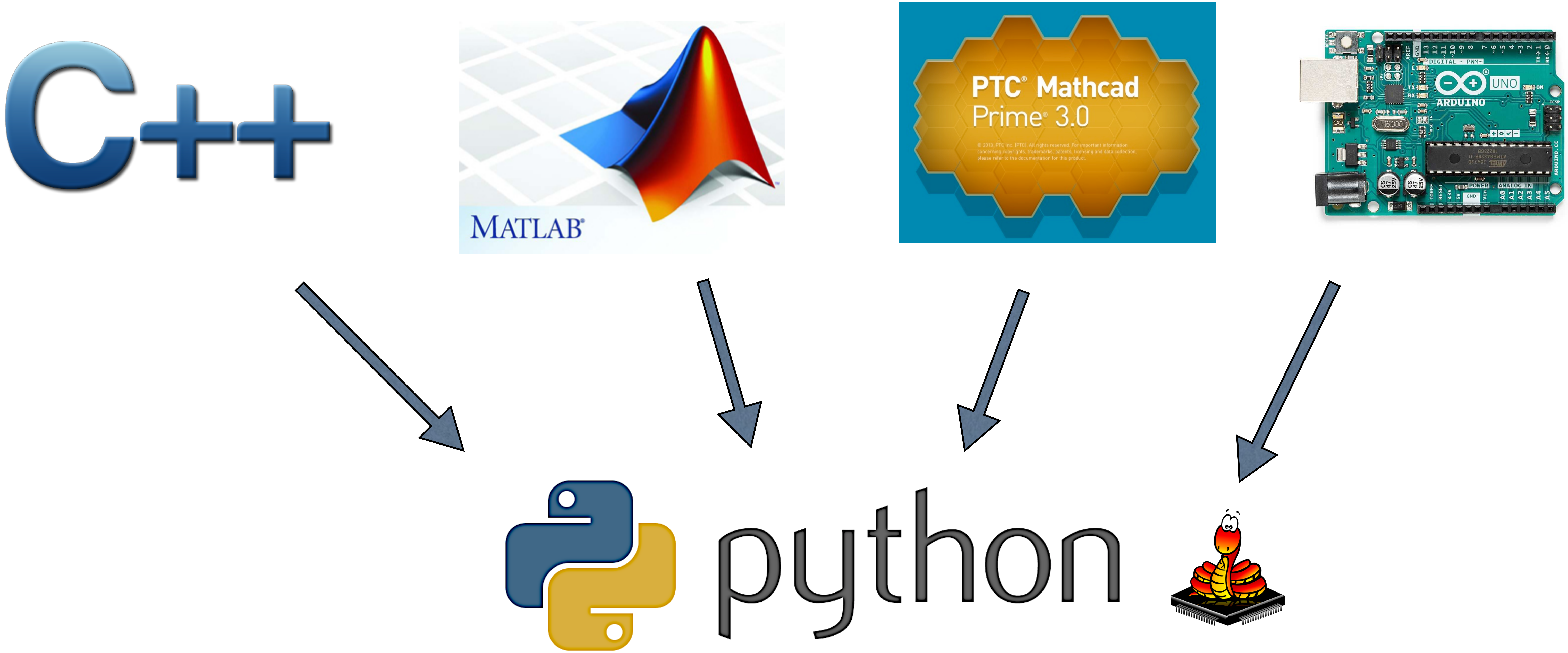
A Unified Lab Environment



Python is the programming framework for Data Acquisition, Data Analysis and Computational Physics in the Advanced Laboratories.

Installing Python

- We use Python 3 (3.8 is installed on lab. computers)
- Unfortunately lab computers are old and we cannot put the most up-to-date Python & Jupyter Notebook / Lab on them.
  - 45 new PCs ordered in the July with 1-week delivery - they arrived lunchtime on monday! We will start install later in the year
- Latest: Python 3.13
- Python on your own laptop:
  - Miniforge for Conda-forge
  - Anaconda Python Distribution (download)
  - Note: there is also
    - Google Colab (link)

How to learn Python for the lab?

- We assume everyone has basic Python from 2Y (Computational Science module) but we recognise there will be a large range of abilities and experiences.
- We used to do intensive introductory sessions but mixed success - students felt too overwhelmed with new material at the start of Stage 3.
- **New approach**:
  - Documentation and examples online
  - HowTo's (developed by a Stage 3 student on a summer internship) in Google Colab (feedback & requests welcome!)
- Informal tutorials can be given as needed:
  - Please coordinate (Class reps) and let me know.

Observations of Student's Code

- Code not commented and poorly structured
  - Long lines extended beyond edge of page ...
    - check that what you submit is not truncated!
    - split code across multiple lines
    - Use Black! (enable with `jupyterlab_code_formatter` extension)
  - Variables poorly named

- Code overly complicated!
  - strive for simplicity and readability
  - Don’t try to automate a complicated analysis so it automatically analyses multiple data files
    - when something goes wrong it is difficult to track down!

Recommendations

"Debugging is twice as hard as writing the code in the first place. Therefore, if you write the code as cleverly as possible, you are, by definition, not smart enough to debug it." (Kernighan's Law)

- Keep it simple!
  - straight-forward easy-to-read and easy-to-follow code is much more understandable and maintainable than complicated code that is difficult to understand.
  - strive to improve - experience and practice is critical!
  - don't be overwhelmed
    - ask for help and advice
  - be critical of results returned by Google (e.g. stackoverflow) etc.
- Python Coding Guidelines: https://physicslabs.ucd.ie/documentation/python/coding_guidelines/

Computer-based DACQ

- You should have one Jupyter Notebook for acquiring and **saving the data**
  - this will probably have some preliminary plots to check data quality, range etc.
  - save the data to a simple text file.

- A second notebook can be used to load the data and perform the analysis/

- If you do not save the data then:
  - you can only analyse the data on the machine it was taken on
  - once the machine (or notebook) is re-started the data is lost
  - even changing a minor aspect such as graph title/label will require taking the data again!

Submission of Notebooks as Appendices

- Do not submit your code as screenshots!
- Do not submit as a page exported a PDF - print to PDF instead.

Data Acquisition Interfacing

- Over to web site....

physicslabs.ucd.ie